Practice workbook · Syntax, logic and runtime errors

Defining logic, syntax and runtime errors, classifying described faults, naming the error from symptoms and giving an example.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-08

1. Question 1 · part b.i · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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1. Question 1 · part b.i · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

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(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part b.i · 3 marks · Working and answer

Working and answer

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2. Question 1 · part b · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part b · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

1 Programmers write code to solve problems.

(a) Give **one** technique that programmers use to make their code easy to read and understand.

(1)

.....

(b) Programs may contain logic, syntax or runtime errors.

A program has three errors.

These may be logic, syntax or runtime errors.

Complete the table by adding a tick (✓) for **each** error description to show whether the error is a logic, syntax or runtime error.

(3)

Error description	Logic	Syntax	Runtime
Divide by 0			
Use x instead of * to multiply			
Subtract 10 from 2 instead of 2 from 10			

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3. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

(c) **Figure 1** shows a programming construct in a flowchart.

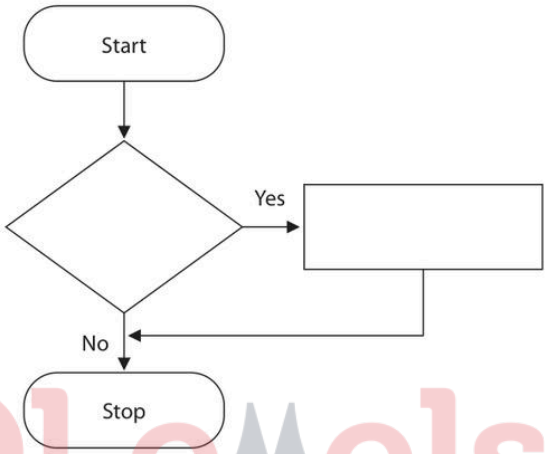


Figure 1

Identify the name of the programming construct.

(1)

- ☒ **A** Array
- ☒ **B** Iteration
- ☒ **C** Operator
- ☒ **D** Selection

(d) Open **Q01d** in the code editor.

The program inputs two numbers, compares them and outputs a suitable message.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01dFINISHED** with the correct file extension for the programming language.

(3)

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3. Question 1 · part b · 3 marks · Working and answer

Working and answer

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4. Question 1 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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4. Question 1 · part e · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part e only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

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(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

4. Question 1 · part e · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b.i only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

☒ **A** Algorithm

☒ **B** Computational thinking

☒ **C** Decomposition

☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

.....

.....

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

.....

.....

(ii) Give **one** example of a runtime error.

(1)

.....

.....

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5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

(ii) Give **one** example of a runtime error.

(1)

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6. Question 2 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

(c) Open **Q02c** in the code editor.

The program generates a random number for the width of a rectangle.

The length of the rectangle is always 4

The program should calculate the perimeter of the rectangle.

The program should display the width, the length and the perimeter of the rectangle.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q02cFINISHED** with the correct file extension for the programming language.

(3)

(d) **Figure 1** shows part of an online form displayed in a browser.

Memorable word		(2-8 letters)
Date	31-01-2019	

Figure 1

Complete the table to show **two** examples of erroneous, **one** example of normal and **one** example of boundary test data for the **memorable word** field.

Each example must be different.

(4)

Erroneous	
Erroneous	
Normal	
Boundary	

(Total for Question 2 = 11 marks)

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6. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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